

Decay Estimates for Weakly Decreasing Solutions to a Weighted Hardy-Sobolev System of Mixed Powers

E. Chernysh

We investigate the decay properties of weakly decreasing solutions to the following weighted Hardy-Sobolev type system:

$$\begin{cases} u(x) = \int_{\mathbb{R}^n} \frac{v(y)^q u(y)^r}{|x-y|^{n-\alpha} |y|^{\sigma_1}} dy & x \in \mathbb{R}^n, \\ v(x) = \int_{\mathbb{R}^n} \frac{u(y)^p v(y)^s}{|x-y|^{n-\alpha} |y|^{\sigma_1}} dy & x \in \mathbb{R}^n \end{cases} \quad (1)$$

in $n \geq 3$ dimensions with $\alpha \in (0, n)$, $0 \leq \sigma_1, \sigma_2 < \alpha$, $p, q > 0$, $r, s \in [0, 1]$ and $pq > 1$. Under appropriate conditions (see [11]) it is equivalent to the fractional system:

$$\begin{cases} (-\Delta)^{\alpha/2} u \equiv \frac{v^q u^r}{|y|^{\sigma_1}} & x \in \mathbb{R}^n, \\ (-\Delta)^{\alpha/2} v \equiv \frac{u^p v^s}{|y|^{\sigma_2}} & x \in \mathbb{R}^n. \end{cases}$$

We derive decay estimates for *weakly monotone decreasing* solutions to (1), i.e. positive functions that satisfy

Definition. A measurable function $f : \mathbb{R}^n \rightarrow (0, \infty)$ is said to be *weakly decreasing* when there exists $C, R > 0$ such that $f(x) \leq Cf(y)$ whenever $|x| \geq |y| \geq R$. The set of all such functions is denoted by $W_m(\mathbb{R}^n)$.

We shall be concerned with weakly monotone decreasing solution pairs to (1). The results we shall state continue to hold true for the more general system:

$$\begin{cases} u(x) = \int_{\mathbb{R}^n} \frac{\varphi(\xi) v(\xi)^q u(\xi)^r + \sum_{j=0}^{n_1} a_j(\xi) u(\xi)^{r_j} v(\xi)^{q_j}}{|x-\xi|^{n-\alpha} |\xi|^{\sigma_1}} d\xi, \\ v(x) = \int_{\mathbb{R}^n} \frac{\psi(\xi) u(\xi)^p v(\xi)^s + \sum_{j=0}^{n_2} b_j(\xi) u(\xi)^{p_j} v(\xi)^{s_j}}{|x-\xi|^{n-\alpha} |\xi|^{\sigma_2}} d\xi \end{cases} \quad x \in \mathbb{R}^n$$

under the appropriate conditions. For ease of notation we handle only system (1)¹. Define:

$$r_0 := \frac{p(\alpha - \sigma_1) + (\alpha - \sigma_2)(1 - r)}{pq - (1 - s)(1 - r)}, \quad s_0 := \frac{q(\alpha - \sigma_2) + (\alpha - \sigma_1)(1 - s)}{pq - (1 - s)(1 - r)}.$$

Further, we set $\mathbf{a} := n/s_0$ and $\mathbf{b} := n/r_0$. We also note that all decay solutions are $W_m(\mathbb{R}^n)$. For an example of a function that is weakly decreasing but not decaying, consider $f(x) := 1/\log x$. In

¹The proofs are very similar to those applicable to the more general system

fact, any function satisfying $0 < m \leq f(x) \leq M$ at infinity is $W_m(\mathbb{R}^n)$.² Unless stated otherwise, we assume that u and v are weakly monotone decreasing solutions, finite for all $x \in \mathbb{R}^n$, to (1).

Theorem 1. *Let (u, v) be a $W_m(\mathbb{R}^n)$ -class solution pair to (1). There holds the following:*

(1) *There exists $C > 0$ such that*

$$u(x) \leq C|x|^{-s_0} \quad \text{and} \quad v(x) \leq C|x|^{-r_0}$$

as $|x| \rightarrow \infty$.

(2) *There exists $C > 0$ and $\delta > 0$ such that*

$$u(x) \geq \frac{C}{(\delta + |x|)^{\min\{n-\alpha, (q+r)(n-\alpha) - (\alpha-\sigma_1)\}}},$$

$$v(x) \geq \frac{C}{(\delta + |x|)^{\min\{n-\alpha, (p+s)(n-\alpha) - (\alpha-\sigma_2)\}}}.$$

as $|x| \rightarrow \infty$.

As well as,

Theorem 2. *Let $(u, v) \in L_{loc}^a(\mathbb{R}^n) \times L_{loc}^b(\mathbb{R}^n)$ be a non-integrable decaying solution pair to (1). Then u and v have the slow rates, i.e.*

$$u(x) \simeq |x|^{-s_0} \quad \text{and} \quad v(x) \simeq |x|^{-r_0}.$$

1. The Proof of Theorem 1

PROOF OF THEOREM 1-(1). Pick $R \gg 0$ and $C > 0$ such that $Cu(x) \leq u(y)$ and $Cv(x) \leq v(y)$ whenever $|x| \geq |y| \geq R$. Given $|x| > 2R$ we define an annulus in $\mathbb{R}^n$:

$$A_x := \{y \in \mathbb{R}^n : R < |y| < |x|\}$$

and write (since $|y| \leq |x|$)

$$u(x) = \int_{\mathbb{R}^n} \frac{v(y)^q u(y)^r}{|x-y|^{n-\alpha} |y|^{\sigma_1}} dy \geq \int_{A_x} \frac{v(y)^q u(y)^r}{|x-y|^{n-\alpha} |y|^{\sigma_1}} dy \geq Cv(x)^q u(x)^r |x|^{-\sigma_1} \int_{A_x} \frac{1}{|x-y|^{n-\alpha}} dy.$$

Noting that $|x-y| \leq |x| + |y| \leq 2|x|$ we find after a possible correction of C :

$$u(x) \geq Cv(x)^q u(x)^r |x|^{-(n-\alpha)-\sigma_1} \text{meas}(A_x) = Cv(x)^q u(x)^r |x|^{\alpha-\sigma_1} \quad (\text{as } |x| \rightarrow \infty)$$

since $\text{meas}(A_x) \simeq |x|^n - R^n \sim |x|^n$. By symmetry there also holds:

$$v(x) \geq Cu(x)^p v(x)^s |x|^{\alpha-\sigma_1} \quad (\text{as } |x| \rightarrow \infty).$$

After “cleaning up” these inequalities we discover that as $|x| \rightarrow \infty$:

$$u(x) \geq Cv(x)^{\frac{q}{1-r}} |x|^{\frac{\alpha-\sigma_1}{1-r}} \quad \text{and} \quad v(x) \geq Cu(x)^{\frac{p}{1-s}} |x|^{\frac{\alpha-\sigma_2}{1-s}}.$$

Using these together for u we obtain the estimate:

$$u(x) \geq C \left(u(x)^{\frac{p}{1-s}} |x|^{\frac{\alpha-\sigma_2}{1-s}} \right)^{\frac{q}{1-r}} |x|^{\frac{\alpha-\sigma_1}{1-r}} = Cu(x)^{\frac{pq}{(1-s)(1-r)}} |x|^{\frac{q(\alpha-\sigma_2)}{(1-s)(1-r)} + \frac{\alpha-\sigma_1}{1-r}}$$

for large x ; thereby establishing

$$u(x)^{1-\frac{pq}{(1-s)(1-r)}} = u(x)^{\frac{(1-s)(1-r)-pq}{(1-s)(1-r)}} \geq C|x|^{\frac{q(\alpha-\sigma_2)+(\alpha-\sigma_1)(1-s)}{(1-s)(1-r)}} \quad (\text{as } |x| \rightarrow \infty).$$

Or, rather,

$$u(x)^{\frac{pq-(1-s)(1-r)}{(1-s)(1-r)}} \leq C|x|^{-\frac{q(\alpha-\sigma_2)+(\alpha-\sigma_1)(1-s)}{(1-s)(1-r)}} \quad (\text{as } |x| \rightarrow \infty).$$

²Consider $f(x) = \sin|x| + 2$.

Taking roots grants us the desired estimate on u . By use of a symmetric argument one may obtain the inequality for v . $\square$

Lemma 3. *Let (u, v) be a weakly monotone decreasing solution pair to (1). There exists $\delta > 0$ and a positive constant C such that*

$$\min \{u(x), v(x)\} \geq \frac{C}{(\delta + |x|)^{n-\alpha}}, \quad \text{as } |x| \rightarrow \infty.$$

PROOF. By the first part of Theorem 1 there exists $R \gg 0$ such that $\max\{u(x), v(x)\} \leq 1$ whenever $|x| \geq R - 1$. Define

$$A := \{y \in \mathbb{R}^n : R - 1 < |y| < R\};$$

there exists $C > 0$ such that

$$C \leq \min \left\{ \int_A \frac{v(y)^q u(y)^r}{|y|^{\sigma_1}} dy, \int_A \frac{u(y)^p v(y)^s}{|y|^{\sigma_2}} dy \right\}$$

where both integrals are finite. Define $\delta := R$ and let $|x| \geq 2R$, by grace of $|x - y| \leq |x| + |y| \leq \delta + |x|$ there holds

$$u(x) \geq \int_A \frac{v(y)^q u(y)^r}{|x - y|^{n-\alpha} |y|^{\sigma_1}} dy \geq \frac{1}{(\delta + |x|)^{n-\alpha}} \int_A \frac{v(y)^q u(y)^r}{|y|^{\sigma_1}} dy \geq \frac{C}{(\delta + |x|)^{n-\alpha}}.$$

One may use a similar argument to show that v also satisfies this inequality. $\square$

PROOF OF THEOREM 1-(2). Here we complete the proof of the first theorem. Define for $x \neq 0$ an open ball:

$$B_x := \left\{ y \in \mathbb{R}^n : |x - y| < \frac{|x|}{2} \right\}$$

and note that $y \in B_x$ may be made arbitrarily large by letting $|x|$ tend to infinity. Hence, by the previous lemma, as $|x| \rightarrow \infty$ one has:

$$\begin{aligned} u(x) &\geq \int_{B_x} \frac{v(y)^q u(y)^r}{|x - y|^{n-\alpha} |y|^{\sigma_1}} dy \geq \frac{C}{(\delta + |x|)^{(q+r)(n-\alpha)}} \int_{B_x} \frac{1}{|x - y|^{n-\alpha} |y|^{\sigma_1}} dy \\ &\geq \frac{C}{(\delta + |x|)^{(q+r)(n-\alpha)+\sigma_1}} \int_{B_x} \frac{1}{|x - y|^{n-\alpha}} dy \\ &\geq \frac{C}{(\delta + |x|)^{(q+r)(n-\alpha)+\sigma_1}} \int_0^{|x|/2} \frac{1}{\rho^{n-\alpha}} \cdot \rho^{n-1} d\rho \\ &\geq \frac{C}{(\delta + |x|)^{(q+r)(n-\alpha)+\sigma_1-\alpha}} \quad (\text{as } |x| \rightarrow \infty). \end{aligned}$$

$\square$

2. The Proof of Theorem 2

We begin this section with a lemma.

Lemma 4. *Let $(u, v) \in L_{loc}^a(\mathbb{R}^n) \times L_{loc}^b(\mathbb{R}^n)$ be a positive non-integrable solution pair to (1). For each $\theta_1 > s_0$ and $\theta_2 > r_0$ there cannot exist $c > 0$ such that*

$$u(x) \leq C |x|^{-\theta_1} \quad \text{or} \quad v(x) \leq C |x|^{-\theta_2}, \quad \text{as } |x| \rightarrow \infty.$$

PROOF. Suppose for a contradiction that $u(x) \leq C|x|^{-\theta_1}$ for some $\theta_1 > s_0$ and some $C > 0$ as $|x| \rightarrow \infty$. For all $R \gg 0$ there then holds by local-integrability of u :

$$\|u\|_{L^a(\mathbb{R}^n)}^a = \int_{|x| < R} u(x)^a dx + \int_{|x| \geq R} u(x)^a dx \leq C_R + C \int_{|x| \geq R} |x|^{-\theta a} dx.$$

Therefore, passing the polar-coordinates yields:

$$\|u\|_{L^a(\mathbb{R}^n)}^a \leq C_R + C \int_R^\infty \rho^{-\theta a + n - 1} d\rho < \infty$$

since $-\theta a + n - 1 < -n + n - 1$. This implies that $u \in L^a(\mathbb{R}^n)$: a contradiction. A symmetric argument shows the inequality in v cannot hold either. This proves the lemma. $\square$

PROOF OF THEOREM 2. By assumption, $u(x) \simeq |x|^{-\theta_1}$ for some $\theta > 0$. By virtue of the previous lemma, we cannot have $\theta_1 > s_0$. To see that $\theta_1 < s_0$ is impossible, suppose for a contradiction that $u(x) \geq c|x|^{-\theta_1}$ for large x . Using the first theorem,

$$c|x|^{-\theta_1} \leq u(x) \leq C|x|^{-s_0} \quad (\text{as } |x| \rightarrow \infty).$$

Which is ridiculous, since $c|x|^{-\theta_1} > C|x|^{-s_0}$ for all x large; this implies that $\theta_1 = s_0$. Following this same procedure, one can show that $v(x) \simeq |x|^{-r_0}$. This concludes the proof. $\square$

3. Integrable Decay Solutions

Throughout this section we assume that u and v are both weakly monotone decreasing and bounded.

Proposition 5. *Let (u, v) be a solution pair to (1). Then u has the fast rate if and only if*

$$\frac{v(x)^q u(x)^r}{|x|^{\sigma_1}} \in L^1(\mathbb{R}^n).$$

PROOF. We fix $R \gg 0$ and consider the decomposition

$$\begin{aligned} u(x) |x|^{n-\alpha} &= \int_{B_R(0)} \frac{v(y)^q u(y)^r |x|^{n-\alpha}}{|y|^{\sigma_1} |x-y|^{n-\alpha}} dy + \int_{B_R(0)^c \setminus B_{\frac{|x|}{2}}(x)} \frac{v(y)^q u(y)^r |x|^{n-\alpha}}{|y|^{\sigma_1} |x-y|^{n-\alpha}} dy \\ &\quad + \int_{B_{\frac{|x|}{2}}(x)} \frac{v(y)^q u(y)^r |x|^{n-\alpha}}{|y|^{\sigma_1} |x-y|^{n-\alpha}} dy; \end{aligned}$$

which we label $I_{1,2,3}$, respectively. The idea is to show that all of the above are bounded in the limit superior, as $|x| \rightarrow \infty$. For the first, we employ dominated convergence to see that, given any $R > 0$,

$$\lim_{|x| \rightarrow \infty} \int_{B_R(0)} \frac{v(y)^q u(y)^r |x|^{n-\alpha}}{|y|^{\sigma_1} |x-y|^{n-\alpha}} dy = \int_{B_R(0)} \frac{v(y)^q u(y)^r}{|y|^{\sigma_1}} dy.$$

To estimate I_2 we simply observe that for $y \in B_R(0)^c \setminus B_{|x|/2}(x)$ one has $|y| \geq |x|/2$ whence

$$I_2 \leq C \int_{B_R(0)^c \setminus B_{\frac{|x|}{2}}(x)} \frac{v(y)^q u(y)^r}{|y|^{\sigma_1}} dy \leq \int_{B_R(0)^c} \frac{v(y)^q u(y)^r}{|y|^{\sigma_1}} dy \xrightarrow{R \rightarrow \infty} 0$$

by dominated convergence. It now only remains to prove that $I_3 < \infty$ for all x large. To see that this is indeed so, write

$$I_3 \leq C \int_{B_{\frac{|x|}{2}}(x)} \frac{v(y)^q u(y)^r}{|y|^{\sigma_1}} dy \leq C'$$

for a constant independent of x and R , by integrability. Hence,

$$\limsup_{|x| \rightarrow \infty} u(x) |x|^{n-\alpha} < \infty$$

implying that u has the fast rate. Conversely, assume that $u \simeq |x|^{-(n-\alpha)}$. Let $R \gg 0$ be very large and consider for $|x| \geq 2R$:

$$u(x) \geq \int_{R < |y| < |x|} \frac{v(y)^q u(y)^r}{|x-y|^{n-\alpha} |y|^{\sigma_1}} dy \geq c |x|^{-(n-\alpha)} \int_{R < |y| < |x|} \frac{v(y)^q u(y)^r}{|y|^{\sigma_1}} dy$$

whence

$$Cu(x) |x|^{n-\alpha} \geq \int_{R < |y| < |x|} \frac{v(y)^q u(y)^r}{|y|^{\sigma_1}} dy, \quad \text{as } |x| \rightarrow \infty.$$

Thus,

$$\limsup_{|x| \rightarrow \infty} \int_{R < |y| < |x|} \frac{v(y)^q u(y)^r}{|y|^{\sigma_1}} dy < \infty.$$

By monotone convergence,

$$\limsup_{|x| \rightarrow \infty} \int_{R < |y| < |x|} \frac{v(y)^q u(y)^r}{|y|^{\sigma_1}} dy = \int_{R < |y|} \frac{v(y)^q u(y)^r}{|y|^{\sigma_1}} dy.$$

Since u and v are bounded, one has

$$\int_{|y| \leq R} \frac{v(y)^q u(y)^r}{|y|^{\sigma_1}} dy < \infty$$

implying that $\frac{v(x)^q u(x)^r}{|x|^{\sigma_1}} dx \in L^1(\mathbb{R}^n)$ as was required. $\square$

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